

Advanced Statistical Methods II

Consolidated Notes: EM & MM Algorithms, and Survival Analysis

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Covers Lectures IV–XI: the EM algorithm (mixtures, incomplete tables, exponential families), the MM algorithm, and survival analysis (hazard, censoring, Kaplan–Meier, Cox regression).

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Part I

The EM Algorithm

1 Missing Data and the Motivation for EM

Throughout applied statistics we frequently face a gap between the data we *wish* we had and the data we *actually* observe: a survey respondent skips a question, a classification label is unknown, a table records only marginal counts, or an experiment is stopped before every unit has failed. Lecture IV opens the course's second block by asking: how should we do maximum likelihood estimation when part of the “natural” data is missing, hidden, or latent?

1.1 A taxonomy of missingness (Rubin, 1976)

Before building algorithms it is worth fixing vocabulary, since the validity of likelihood-based methods for incomplete data depends on *why* data are missing. Let $Y = (Y_{\text{obs}}, Y_{\text{mis}})$ be the full data and R the missingness indicator ($R_{ij} = 1$ if Y_{ij} is observed). Rubin's classification is:

- **MCAR** (missing completely at random): $P(R \mid Y_{\text{obs}}, Y_{\text{mis}}) = P(R)$ – missingness is unrelated to any data values.
- **MAR** (missing at random): $P(R \mid Y_{\text{obs}}, Y_{\text{mis}}) = P(R \mid Y_{\text{obs}})$ – missingness may depend on what is observed, but not on the missing values themselves once we condition on what is observed.
- **MNAR** (missing not at random): missingness depends on the unobserved values even after conditioning on what is observed.

Key Idea

When missingness is MAR (and the parameters governing the missingness mechanism are distinct from θ , the parameter of interest – the *ignorability* condition), likelihood-based inference can proceed by simply maximizing the observed-data likelihood $L(\theta; Y_{\text{obs}})$, ignoring the mechanism that produced the missingness. Latent-variable problems (mixture labels, censoring indicators whose distribution does not involve θ) are typically treated as ignorable in this sense, which is what licenses the entire EM machinery that follows.

1.2 Two likelihoods

Let Y denote the observed data, Z the missing/latent data, and θ the parameter vector of interest. Two log-likelihoods matter:

- **Complete-data log-likelihood:** $\ell_c(\theta; Y, Z) = \log p(Y, Z \mid \theta)$. This is the likelihood we would use if Z were observed alongside Y ; it is typically simple (often exponential-family, separable across parameters).
- **Observed-data log-likelihood:** $\ell(\theta; Y) = \log p(Y \mid \theta) = \log \sum_z p(Y, z \mid \theta)$ (or an integral for continuous Z). This is what we actually want to maximize, but the log of a sum/integral generally destroys closed-form maximization.

Key Idea

Why EM helps: $\ell_c(\theta; Y, Z)$ is typically linear in the (unobserved) complete-data sufficient statistics, so if we could plug in a good guess for those statistics, maximization would again be easy. EM's idea is to plug in the *conditional expectation* of ℓ_c given the observed data and the current parameter guess, then maximize that expectation – repeating until convergence.

2 The General EM Algorithm

2.1 The Q -function and the two steps

At iteration t , given current estimate $\theta^{(t)}$, define the **Q -function**

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\ell_c(\theta; Y, Z) \mid Y] = \sum_z \ell_c(\theta; Y, z) p(z \mid Y, \theta^{(t)}).$$

This is an expectation over Z taken with respect to its conditional (posterior) distribution given Y *under the current parameter* $\theta^{(t)}$ – not under the parameter θ being optimized. This distinction is the single most commonly tested conceptual point: $\theta^{(t)}$ only fixes the distribution used to average; θ is the free variable maximized over afterward.

The EM Algorithm

E-step (Expectation):

compute the conditional distribution $p(Z \mid Y, \theta^{(t)})$, and hence $Q(\theta \mid \theta^{(t)})$ – or, more practically, the conditional expectations of the sufficient statistics appearing in ℓ_c .

M-step (Maximization):

update $\theta^{(t+1)} = \arg \max_{\theta} Q(\theta \mid \theta^{(t)})$.

Repeat until the parameter estimates (or the observed log-likelihood) stop changing appreciably.

2.2 The ascent property

Theorem 2.1 (EM never decreases the observed likelihood). *If the M-step is exact (i.e. $Q(\theta^{(t+1)} \mid \theta^{(t)}) \geq Q(\theta^{(t)} \mid \theta^{(t)})$), then*

$$\ell(\theta^{(t+1)}; Y) \geq \ell(\theta^{(t)}; Y) \quad \text{for every iteration } t.$$

Proof. Write, for discrete Z with posterior weights $r_z^{(t)} := p(z \mid Y, \theta^{(t)})$,

$$\ell(\theta; Y) = \log \sum_z p(Y, z \mid \theta) = \log \sum_z r_z^{(t)} \frac{p(Y, z \mid \theta)}{r_z^{(t)}}.$$

Since log is concave and the $r_z^{(t)}$ sum to 1, Jensen's inequality gives

$$\ell(\theta; Y) \geq \sum_z r_z^{(t)} \log \frac{p(Y, z \mid \theta)}{r_z^{(t)}} = \underbrace{\sum_z r_z^{(t)} \log p(Y, z \mid \theta)}_{= Q(\theta \mid \theta^{(t)})} - \sum_z r_z^{(t)} \log r_z^{(t)}.$$

Call this lower bound $B(\theta \mid \theta^{(t)}) := Q(\theta \mid \theta^{(t)}) - H(\theta^{(t)})$, where $H(\theta^{(t)}) := \sum_z r_z^{(t)} \log r_z^{(t)}$ does *not* depend on θ . At $\theta = \theta^{(t)}$, $p(Y, z \mid \theta^{(t)})/r_z^{(t)} = p(Y, z \mid \theta^{(t)})/p(z \mid Y, \theta^{(t)}) = p(Y \mid \theta^{(t)})$, a constant in z , so Jensen's inequality holds with *equality*: $B(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)}; Y)$. Hence

$$\begin{aligned} \ell(\theta^{(t+1)}; Y) &\geq B(\theta^{(t+1)} \mid \theta^{(t)}) = Q(\theta^{(t+1)} \mid \theta^{(t)}) - H(\theta^{(t)}) \\ &\geq Q(\theta^{(t)} \mid \theta^{(t)}) - H(\theta^{(t)}) = B(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)}; Y). \end{aligned}$$

The middle inequality uses the M-step's defining property. ■

□

Remark 2.2. This proof is the discrete- Z version of the standard KL-divergence argument: writing $\ell(\theta; Y) = Q(\theta \mid \theta^{(t)}) - H(\theta \mid \theta^{(t)})$ with $H(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\log p(Z \mid Y, \theta) \mid Y]$, Gibbs' inequality (nonnegativity of KL divergence) gives $H(\theta \mid \theta^{(t)}) \leq H(\theta^{(t)} \mid \theta^{(t)})$ for every θ , with equality at $\theta = \theta^{(t)}$. Combined with the M-step inequality on Q , the two facts add to give Theorem 2.1. This is exactly the majorization argument that reappears in Section II as a special case of the general MM principle.

Key Idea

Consequences and caveats (frequently tested):

- EM converges to a *stationary point* of the observed likelihood (under mild regularity), not necessarily the global maximum.

- Multimodal likelihoods (e.g., normal mixtures) mean different starting values $\theta^{(0)}$ can lead to different local maxima – hence the recommendation of multiple random starts.
- The sequence $\ell(\theta^{(t)}; Y)$ is monotone nondecreasing and, if bounded above, converges; this gives a natural stopping rule: stop when $\ell(\theta^{(t+1)}; Y) - \ell(\theta^{(t)}; Y)$ or $\|\theta^{(t+1)} - \theta^{(t)}\|$ is smaller than a preset tolerance.

2.3 Common misconceptions

1. “The E-step chooses the most likely value of every missing quantity.” False – the E-step *averages* over the conditional distribution of the missing/latent quantities; it does not make hard assignments.
2. “The M-step maximizes the observed log-likelihood directly.” False – it maximizes the surrogate Q , which is constructed precisely so that improving it improves the observed likelihood.
3. “Monotone likelihood means we’ve found the global maximum.” False – only a stationary point is guaranteed.
4. “EM automatically gives standard errors.” False – see Section 8.2.
5. “EM is only useful when literal database entries are missing.” False – latent class labels, random effects, and censoring indicators all create EM problems even when no literal blank cells exist.

3 Worked Example I: A Two-Component Poisson Mixture

This example (Lecture IV) contains, in miniature, every idea used later for normal mixtures.

3.1 Setup

Items are produced by one of two machines, with the machine label unrecorded. For item i , let $Z_i \in \{1, 2\}$ with $P(Z_i = 1) = \pi$, and $Y_i \mid Z_i = k \sim \text{Poisson}(\lambda_k)$. We observe only $y_1, \dots, y_n$. The observed-data log-likelihood is

$$\ell(\pi, \lambda_1, \lambda_2) = \sum_{i=1}^n \log \left\{ \pi f_1(y_i; \lambda_1) + (1 - \pi) f_2(y_i; \lambda_2) \right\}, \quad f_k(y; \lambda_k) = \frac{e^{-\lambda_k} \lambda_k^y}{y!},$$

whose log-of-a-sum structure again prevents a closed-form MLE.

3.2 Complete-data likelihood and its MLEs

With $Z_{i1} = \mathbf{1}(Z_i = 1)$, $Z_{i2} = 1 - Z_{i1}$, the complete-data log-likelihood is (up to an additive constant)

$$\ell_c(\theta) = \sum_{i=1}^n \left[Z_{i1} \{ \log \pi + y_i \log \lambda_1 - \lambda_1 \} + Z_{i2} \{ \log(1 - \pi) + y_i \log \lambda_2 - \lambda_2 \} \right].$$

Differentiating the π -terms, $\sum_i Z_{i1}/\pi = \sum_i Z_{i2}/(1 - \pi)$, and using $Z_{i1} + Z_{i2} = 1$ gives $\hat{\pi} = \frac{1}{n} \sum_i Z_{i1}$. Differentiating the λ_1 -terms, $\sum_i Z_{i1}(y_i/\lambda_1 - 1) = 0$ gives $\hat{\lambda}_1 = \sum_i Z_{i1} y_i / \sum_i Z_{i1}$, and symmetrically $\hat{\lambda}_2 = \sum_i Z_{i2} y_i / \sum_i Z_{i2}$. The only obstacle is that the Z_i ’s are unobserved; EM replaces them by their conditional expectations.

3.3 E-step and M-step

The **responsibility** of Machine 1 for observation i at iteration t is

$$\tau_i^{(t)} = P(Z_i = 1 \mid Y_i = y_i, \theta^{(t)}) = \frac{\pi^{(t)} f_1(y_i; \lambda_1^{(t)})}{\pi^{(t)} f_1(y_i; \lambda_1^{(t)}) + (1 - \pi^{(t)}) f_2(y_i; \lambda_2^{(t)})}.$$

The M-step substitutes $\tau_i^{(t)}$ for Z_{i1} (and $1 - \tau_i^{(t)}$ for Z_{i2}) in the complete-data formulas:

$$\pi^{(t+1)} = \frac{1}{n} \sum_i \tau_i^{(t)}, \quad \lambda_1^{(t+1)} = \frac{\sum_i \tau_i^{(t)} y_i}{\sum_i \tau_i^{(t)}}, \quad \lambda_2^{(t+1)} = \frac{\sum_i (1 - \tau_i^{(t)}) y_i}{\sum_i (1 - \tau_i^{(t)})}.$$

3.4 Numerical illustration

Data: 0, 1, 2, 5, 6; start $\pi^{(0)} = 0.5$, $\lambda_1^{(0)} = 1$, $\lambda_2^{(0)} = 5$.

y_i	0	1	2	5	6
$\tau_i^{(0)}$	0.982	0.916	0.686	0.017	0.003

Small counts are assigned mostly to Machine 1, large counts to Machine 2. With $\sum_i \tau_i^{(0)} \approx 2.605$ and $\sum_i \tau_i^{(0)} y_i \approx 2.395$: $\pi^{(1)} \approx 0.521$, $\lambda_1^{(1)} \approx 0.919$, $\lambda_2^{(1)} \approx 4.845$. Iterating further, $(\pi^{(2)}, \lambda_1^{(2)}, \lambda_2^{(2)}) \approx (0.515, 0.903, 4.816)$ and $(\pi^{(3)}, \lambda_1^{(3)}, \lambda_2^{(3)}) \approx (0.511, 0.894, 4.793)$, with $\ell(\theta^{(t)})$ increasing monotonically: $-10.318 \rightarrow -10.298 \rightarrow -10.296 \rightarrow -10.295$.

4 Worked Example II: Poisson Data with Missing Counts

Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\lambda)$, but only the first r counts are observed and $m = n - r$ are missing, with $S_{\text{obs}} = \sum_{i=1}^r X_i$. (Direct maximization is trivial here – $\hat{\lambda} = S_{\text{obs}}/r$ – but the example cleanly displays every EM step algebraically.)

The complete-data log-likelihood is $\ell_c(\lambda) = (\sum_{i=1}^n X_i) \log \lambda - n\lambda + C$, so the only missing sufficient statistic is the sum of the missing counts. Since each missing X_i has $\mathbb{E}[X_i | \theta^{(t)}] = \lambda^{(t)}$, the E-step gives expected missing total $m\lambda^{(t)}$, and the M-step is

$$\lambda^{(t+1)} = \frac{S_{\text{obs}} + m\lambda^{(t)}}{n}.$$

With $n = 10, r = 6, m = 4, S_{\text{obs}} = 18, \lambda^{(0)} = 5$: $\lambda^{(1)} = 3.8$, $\lambda^{(2)} = 3.32$, $\lambda^{(3)} = 3.128, \dots$. At the fixed point, $\lambda^* = (S_{\text{obs}} + m\lambda^*)/n \Rightarrow (n - m)\lambda^* = S_{\text{obs}} \Rightarrow \lambda^* = S_{\text{obs}}/r$, exactly the observed-data MLE $\bar{X}_{\text{obs}}$.

Key Idea

The missing counts here carry *no extra information* about λ (all observations, missing or not, are iid), so EM converges to the ordinary observed-data MLE. In richer models the same architecture survives even when the algebra is far less trivial: (1) identify missing sufficient statistics; (2) take their conditional expectation given current parameters; (3) plug into the complete-data estimation formula; (4) iterate.

5 EM for Normal Mixtures

5.1 The finite mixture model

For K hidden components, each unit $i = 1, \dots, n$ draws a label $Z_i \in \{1, \dots, K\}$ with $P(Z_i = k) = \pi_k$ ($\pi_k > 0$, $\sum_k \pi_k = 1$), then $Y_i | Z_i = k \sim f_k(y; \theta_k)$. Marginalizing Z_i ,

$$f(y; \theta) = \sum_{k=1}^K \pi_k f_k(y; \theta_k).$$

For a **normal mixture**, $Y_i | Z_i = k \sim N(\mu_k, \sigma_k^2)$, so $f(y_i; \theta) = \sum_k \pi_k \phi(y_i; \mu_k, \sigma_k^2)$, with $\theta = (\pi_k, \mu_k, \sigma_k^2)_{k=1}^K$.

Key Idea

A mixture of normals can produce skewness, multimodality, and heavy-tailed shapes even though every individual component is Gaussian – “mixture” signals a hidden per-observation label we do not observe.

5.2 Why direct maximization is awkward

Without labels, $\ell(\theta) = \sum_{i=1}^n \log \{ \sum_k \pi_k \phi(y_i; \mu_k, \sigma_k^2) \}$. The log-of-a-sum: (i) is usually non-concave in θ ; (ii) may have multiple local maxima; (iii) admits no closed-form maximizer. This motivates introducing latent labels and EM.

5.3 Complete-data likelihood and the E-step

With indicators $Z_{ik} = \mathbf{1}(Z_i = k)$, the complete-data log-likelihood (dropping additive constants) is

$$\ell_c(\theta) = \sum_{i=1}^n \sum_{k=1}^K Z_{ik} \left[\log \pi_k - \frac{1}{2} \log \sigma_k^2 - \frac{(y_i - \mu_k)^2}{2\sigma_k^2} \right].$$

Since Z_{ik} is 0/1, its conditional expectation is a conditional probability, called the **responsibility**:

$$\tau_{ik}^{(t)} := \mathbb{E}_{\theta^{(t)}}[Z_{ik} | y_i] = P_{\theta^{(t)}}(Z_i = k | Y_i = y_i) = \frac{\pi_k^{(t)} \phi(y_i; \mu_k^{(t)}, (\sigma_k^2)^{(t)})}{\sum_{\ell=1}^K \pi_\ell^{(t)} \phi(y_i; \mu_\ell^{(t)}, (\sigma_\ell^2)^{(t)})},$$

derived directly from Bayes' rule: $P(Z_i = k | y_i, \theta^{(t)}) = \frac{P(Z_i = k | \theta^{(t)}) f(y_i | Z_i = k, \theta^{(t)})}{f(y_i | \theta^{(t)})}$. Properties:

$0 \leq \tau_{ik}^{(t)} \leq 1$ and $\sum_k \tau_{ik}^{(t)} = 1$ for every i . Substituting $\tau_{ik}^{(t)}$ for Z_{ik} gives

$$Q(\theta | \theta^{(t)}) = \sum_{i=1}^n \sum_{k=1}^K \tau_{ik}^{(t)} \left[\log \pi_k - \frac{1}{2} \log \sigma_k^2 - \frac{(y_i - \mu_k)^2}{2\sigma_k^2} \right] + \text{const.}$$

Soft clustering

A hard clustering method sends each Y_i to exactly one component. EM instead returns a full probability vector $(\tau_{i1}^{(t)}, \dots, \tau_{iK}^{(t)})$ – observations near one component get responsibility close to $(1, 0, \dots)$; observations in an overlap region get genuinely fractional, in-between probabilities. This is why EM for mixtures is called *soft* clustering.

5.4 M-step: weighted MLE updates

Maximizing Q subject to $\sum_k \pi_k = 1$ (full derivations in Section 8.1) gives

$$\pi_k^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \tau_{ik}^{(t)}, \quad \mu_k^{(t+1)} = \frac{\sum_i \tau_{ik}^{(t)} y_i}{\sum_i \tau_{ik}^{(t)}}, \quad (\sigma_k^2)^{(t+1)} = \frac{\sum_i \tau_{ik}^{(t)} (y_i - \mu_k^{(t+1)})^2}{\sum_i \tau_{ik}^{(t)}}.$$

Common exam error

The variance update must use the *updated* mean $\mu_k^{(t+1)}$, not the old mean $\mu_k^{(t)}$ – using the wrong mean is one of the most frequent mistakes in mixture-EM derivations.

5.5 The full algorithm

EM for Univariate Normal Mixtures

1. Choose starting values $\theta^{(0)}$.
2. For $t = 0, 1, 2, \dots$: (a) E-step – compute $\tau_{ik}^{(t)}$ for all i, k ; (b) M-step – update π_k, μ_k, σ_k^2 ; (c) evaluate $\ell(\theta^{(t)}) = \sum_i \log \{ \sum_k \pi_k^{(t)} \phi(y_i; \mu_k^{(t)}, (\sigma_k^2)^{(t)}) \}$.
3. Stop when the parameter change or likelihood improvement is below a tolerance.

5.6 Worked numerical example (two components)

Data $y = (-2, -1.5, -1, 1, 1.5, 2)$, $K = 2$, common known variance $\sigma_1^2 = \sigma_2^2 = 1$; initialize $\pi^{(0)} = 0.5$, $\mu_1^{(0)} = -1$, $\mu_2^{(0)} = 1$. Since $\pi^{(0)} = 0.5$, symmetric points near -1 get $\tau_{i1}^{(0)} \approx 1$ and points near $+1$ get $\tau_{i1}^{(0)} \approx 0$:

$$\tau_{\cdot 1}^{(0)} \approx (0.998, 0.982, 0.881, 0.119, 0.018, 0.002).$$

M-step: $\pi^{(1)} = \frac{1}{6}(2.998 + 2.982 + \dots) = 0.500$ and

$$\mu_1^{(1)} = \frac{(0.998)(-2) + (0.982)(-1.5) + (0.881)(-1) + (0.119)(1) + (0.018)(1.5) + (0.002)(2)}{3.000} \approx -1.49,$$

with $\mu_2^{(1)} \approx +1.49$ by symmetry. Repeating converges to $\hat{\mu}_1 = -1.5$, $\hat{\mu}_2 = 1.5$, $\hat{\pi} = 0.5$, matching the “obvious” hard-clustering answer here because the groups are well separated.

Key Idea

When clusters are well separated, EM’s soft assignments quickly sharpen toward hard 0/1 labels; the interesting/hard cases are when clusters overlap.

5.7 Practical issues

Initialization and label switching. EM only guarantees the observed log-likelihood never decreases; the surface can have several local maxima, and it is invariant under permuting (“relabeling”) the K components – **label switching**. Practical starting strategies: random initial allocations, quantile-based splitting into K groups, or several random starts keeping the run with the largest final likelihood.

Choosing K . A larger K can never decrease the fitted likelihood but risks overfitting. Common criteria penalize model complexity:

$$\text{AIC} = -2\hat{\ell} + 2p, \quad \text{BIC} = -2\hat{\ell} + p \log n,$$

where p is the number of free parameters ($p = 3K - 1$ for a univariate normal K -mixture) and $\hat{\ell}$ is the maximized log-likelihood; smaller AIC/BIC is preferred, with BIC penalizing complexity more heavily for large n . Domain knowledge, held-out validation, and interpretability of the resulting components are equally important in practice.

Classic trick question – identical components

Suppose $\mu_1 = \mu_2 = \mu$ and $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (two components sharing mean and variance). Then in the E-step formula $\phi(y_i; \mu_k^{(t)}, (\sigma_k^2)^{(t)})$ is literally the same number for $k = 1, 2$, so it cancels between numerator and denominator: $\tau_{i1}^{(t)} = \pi_1^{(t)}$ for *every* i , regardless of y_i ! The data carry no information about how to split observations between identical components. The M-step re-estimates π_k via its own weighted average (a fixed point immediately, since $\tau_{ik}^{(t)} = \pi_k^{(t)}$ already), while μ, σ^2 converge to the ordinary pooled sample mean and variance – equivalent to fitting a single normal to all the data, since the two components are unidentifiable under this restriction.

6 Solved Problem: EM for Bivariate Normal with One Missing Value

Setup. Suppose $(x_1, y_1), \dots, (x_{n-1}, y_{n-1}), (x_n, *)$ are observations from a Bivariate Normal distribution, $*$ denoting a missing value, with $x_n = \bar{x}_{n-1} = \frac{1}{n-1} \sum_{i=1}^{n-1} x_i$. Using the MLEs from the first $n - 1$ complete observations as initial values, we show the EM algorithm converges in exactly two cycles.

Notation. Let $m = n - 1$; define sample moments $\bar{x}, \bar{y}, s_{xx}, s_{yy}, s_{xy}$ from the first m observations in the usual way, so $\mu_x^{(0)} = \bar{x}$, $\mu_y^{(0)} = \bar{y}$, $\sigma_{xx}^{(0)} = s_{xx}$, $\sigma_{yy}^{(0)} = s_{yy}$, $\sigma_{xy}^{(0)} = s_{xy}$. Crucially $x_n = \bar{x} = \mu_x^{(0)}$: the missing observation’s known x -coordinate exactly equals the current estimate of μ_x .

E-step, cycle 1. For bivariate normal data, $Y | X = x \sim N(\mu_y + \frac{\sigma_{xy}}{\sigma_{xx}}(x - \mu_x), \sigma_{yy} - \frac{\sigma_{xy}^2}{\sigma_{xx}})$. Since $x_n = \mu_x^{(0)}$, the correction term vanishes:

$$\hat{y}_n^{(1)} = \mathbb{E}_{\theta^{(0)}}(Y_n | X_n = x_n) = \mu_y^{(0)} = \bar{y}, \quad \mathbb{E}_{\theta^{(0)}}(Y_n^2 | x_n) = \bar{y}^2 + \tau^2, \quad \tau^2 := s_{yy} - \frac{s_{xy}^2}{s_{xx}}.$$

The second (conditional) moment is essential – the variance update needs $\mathbb{E}(Y_n^2 | x_n)$, not just $\hat{y}_n^2$.

M-step, cycle 1. Treating $\hat{y}_n^{(1)} = \bar{y}$ as the imputed value but using the correct second moment for variance terms:

$$\begin{aligned} \mu_x^{(1)} &= \frac{m\bar{x} + x_n}{n} = \bar{x}, & \mu_y^{(1)} &= \frac{m\bar{y} + \bar{y}}{n} = \bar{y}, \\ \sigma_{xx}^{(1)} &= \frac{m}{n} s_{xx} \quad (\text{since } x_n - \bar{x} = 0), & \sigma_{xy}^{(1)} &= \frac{m}{n} s_{xy}, \\ \sigma_{yy}^{(1)} &= \frac{1}{n} \left\{ m s_{yy} + s_{yy} - \frac{s_{xy}^2}{s_{xx}} \right\} = s_{yy} - \frac{1}{n} \frac{s_{xy}^2}{s_{xx}}. \end{aligned}$$

Why cycle 2 changes nothing. Since $\mu_x^{(1)} = \bar{x}$ again, the same cancellation recurs in the second E-step: $\mathbb{E}_{\theta^{(1)}}(Y_n | x_n) = \mu_y^{(1)} = \bar{y}$. The conditional variance under $\theta^{(1)}$ is

$$\sigma_{yy}^{(1)} - \frac{(\sigma_{xy}^{(1)})^2}{\sigma_{xx}^{(1)}} = \left(s_{yy} - \frac{1}{n} \frac{s_{xy}^2}{s_{xx}} \right) - \frac{(\frac{m}{n} s_{xy})^2}{\frac{m}{n} s_{xx}} = \left(s_{yy} - \frac{1}{n} \frac{s_{xy}^2}{s_{xx}} \right) - \frac{m}{n} \frac{s_{xy}^2}{s_{xx}}.$$

Since $m = n - 1$, $\frac{1}{n} + \frac{m}{n} = 1$, so this equals $s_{yy} - \frac{s_{xy}^2}{s_{xx}} = \tau^2$ – identical to cycle 1’s conditional variance. Hence the second E-step reproduces exactly the same conditional moments, so $\theta^{(2)} = \theta^{(1)}$.

Key Idea

$\theta^{(0)} \rightarrow \theta^{(1)} \rightarrow \theta^{(2)} = \theta^{(1)}$: EM reaches a fixed point immediately, i.e. converges in two cycles. The result depends crucially on the special alignment $x_n = \bar{x}$; without it, the correction term would not vanish and further iterations would generally be needed.

Common mistake

Never replace Y_n only by its conditional mean when estimating variances – EM requires the conditional *second moment* $\mathbb{E}(Y_n^2 | x_n) = \hat{y}_n^2 + (\text{conditional variance})$, not just $\hat{y}_n^2$.

7 EM for Incomplete Tables

7.1 Complete multinomial counts (baseline)

If units fall into K categories with $P(\text{category } k) = p_k$ and all cell counts N_k ($\sum_k N_k = N$) are observed, $(N_1, \dots, N_K) \sim \text{Multinomial}(N; p_1, \dots, p_K)$, and $\ell_c(p) = \sum_k N_k \log p_k$. Maximizing subject to $\sum_k p_k = 1$ via a Lagrange multiplier $L = \sum_k N_k \log p_k + \lambda(1 - \sum_k p_k)$: setting $\partial L / \partial p_k = N_k / p_k - \lambda = 0$ gives $p_k = N_k / \lambda$; summing and using the constraint gives $\lambda = N$, so $\hat{p}_k = N_k / N$.

7.2 A simple incomplete table (one ambiguous group)

Three categories $p = (p_1, p_2, p_3)$; observed: n_1, n_2, n_3 exact counts, plus m units known only to be “category 1 or 2”. Total $N = n_1 + n_2 + n_3 + m$. The observed-data likelihood is $L(p; Y) \propto p_1^{n_1} p_2^{n_2} p_3^{n_3} (p_1 + p_2)^m$.

Let U be the number of the m ambiguous units truly in category 1. If U were known, complete counts would be $N_1 = n_1 + U$, $N_2 = n_2 + (m - U)$, $N_3 = n_3$. Given membership in $\{1, 2\}$, the conditional probability of truly being category 1 is $p_1 / (p_1 + p_2)$, so $U | Y, p^{(t)} \sim \text{Binomial}(m, p_1^{(t)} / (p_1^{(t)} + p_2^{(t)}))$, giving expected counts

$$\tilde{N}_1^{(t)} = n_1 + m \frac{p_1^{(t)}}{p_1^{(t)} + p_2^{(t)}}, \quad \tilde{N}_2^{(t)} = n_2 + m \frac{p_2^{(t)}}{p_1^{(t)} + p_2^{(t)}}, \quad \tilde{N}_3^{(t)} = n_3.$$

The M-step is the ordinary multinomial MLE applied to expected counts: $p_k^{(t+1)} = \tilde{N}_k^{(t)} / N$.

Example 7.1 (Fully worked). $n_1 = 18, n_2 = 12, n_3 = 20, m = 10, N = 60$, start $p^{(0)} = (1/3, 1/3, 1/3)$.

Iter. t	$\hat{U}^{(t)}$	$p_1^{(t+1)}$	$p_2^{(t+1)}$	$p_3^{(t+1)}$
0	5.0000	0.3833	0.2833	0.3333
1	5.7500	0.3958	0.2708	0.3333
2	5.9375	0.3990	0.2677	0.3333
3	5.9844	0.3997	0.2669	0.3333

The sequence converges to $(0.4000, 0.2667, 0.3333)$, with the observed log-likelihood increasing monotonically. *Sanity check:* category 3 is fully observed so $\hat{p}_3 = 20/60 = 1/3$ directly; among the exactly-observed “1 or 2” units the split ratio is $n_1/(n_1 + n_2) = 0.6$, and since $\hat{p}_1 + \hat{p}_2 = 1 - \hat{p}_3 = 40/60$, we get $\hat{p}_1 = 0.6 \times \frac{40}{60} = 0.4$ and $\hat{p}_2 = 0.4 \times \frac{40}{60} = 0.2667$ – matching the EM limit exactly, confirming EM reaches the closed-form MLE via repeated soft completions.

7.3 Solved problem: overlapping ambiguous categories

Three categories; exact counts n_1, n_2, n_3 ; m_{12} units known only as “1 or 2”; m_{23} units known only as “2 or 3”. At iteration t , split each ambiguous group using current probabilities:

$$\tilde{u}_{12}^{(t)} = m_{12} \frac{p_1^{(t)}}{p_1^{(t)} + p_2^{(t)}}, \quad \tilde{u}_{23}^{(t)} = m_{23} \frac{p_2^{(t)}}{p_2^{(t)} + p_3^{(t)}}.$$

Expected complete counts:

$$\tilde{N}_1^{(t)} = n_1 + \tilde{u}_{12}^{(t)}, \quad \tilde{N}_2^{(t)} = n_2 + \{m_{12} - \tilde{u}_{12}^{(t)}\} + \tilde{u}_{23}^{(t)}, \quad \tilde{N}_3^{(t)} = n_3 + \{m_{23} - \tilde{u}_{23}^{(t)}\}.$$

The trick to notice

Category 2 receives expected contributions from *two* different ambiguous groups simultaneously (leftover from m_{12} and leftover from m_{23}) – easy to miss under time pressure. The M-step is still $p_k^{(t+1)} = \tilde{N}_k^{(t)} / N$ with $N = n_1 + n_2 + n_3 + m_{12} + m_{23}$.

7.4 From one-way to two-way (contingency) tables

Let $p_{ij} = P(\text{row } i, \text{col } j)$, complete counts n_{ij} : $\ell_c(p) = \sum_{ij} n_{ij} \log p_{ij}$, $\hat{p}_{ij} = n_{ij}/N$. A record may be incomplete in several ways: *row-only* (row observed, column missing, count a_i), *column-only* (count b_j), or *neither observed* (count r).

Row-only records. Given row i , $P(\text{col } j \mid \text{row } i) = p_{ij}/p_{i+}$ where $p_{i+} = \sum_j p_{ij}$; expected contribution to cell (i, j) : $a_i p_{ij}^{(t)} / p_{i+}^{(t)}$.

Column-only records. Analogously, $b_j p_{ij}^{(t)} / p_{+j}^{(t)}$, with $p_{+j} = \sum_i p_{ij}$.

Fully missing records. No conditioning information, so the expected contribution is simply $r p_{ij}^{(t)}$ – these records are split in exactly the current joint proportions and mostly affect the normalizing total N .

Expected completed two-way table

$$\tilde{n}_{ij}^{(t)} = c_{ij} + a_i \frac{p_{ij}^{(t)}}{p_{i+}^{(t)}} + b_j \frac{p_{ij}^{(t)}}{p_{+j}^{(t)}} + r p_{ij}^{(t)}, \quad p_{ij}^{(t+1)} = \frac{\tilde{n}_{ij}^{(t)}}{N},$$

where c_{ij} is the fully observed count in cell (i, j) and $N = \sum_{ij} c_{ij} + \sum_i a_i + \sum_j b_j + r$.

Example 7.2 (Fully worked 2×2 table). Complete counts $\begin{pmatrix} 12 & 8 \\ 6 & 14 \end{pmatrix}$; $a_1 = 5, a_2 = 3, b_1 = 4, b_2 = 2, r = 0$; $N = 54$. Starting from the normalized complete table, $p^{(0)} = \begin{pmatrix} 0.30 & 0.20 \\ 0.15 & 0.35 \end{pmatrix}$, margins $p_{1+}^{(0)} = p_{2+}^{(0)} = 0.50$, $p_{+1}^{(0)} = 0.45$, $p_{+2}^{(0)} = 0.55$. Row 1's $a_1 = 5$ missing-column records split as $5 \times \frac{0.30}{0.50} = 3.00$ into $(1, 1)$ and $5 \times \frac{0.20}{0.50} = 2.00$ into $(1, 2)$; analogous splits apply to a_2, b_1, b_2 . Combining all four contributions:

$$\tilde{n}^{(0)} \approx \begin{pmatrix} 17.67 & 10.73 \\ 8.23 & 17.37 \end{pmatrix}, \quad p^{(1)} \approx \begin{pmatrix} 0.3272 & 0.1988 \\ 0.1525 & 0.3217 \end{pmatrix}.$$

Further iterations refine the table using updated margins $p_{i+}^{(1)}, p_{+j}^{(1)}$ at each step.

8 The General Recipe: Missing-Data Likelihoods via Exponential Families

Many complete-data log-likelihoods take the canonical exponential-family form

$$\ell_c(\theta; Y, Z) = \eta(\theta)^\top S(Y, Z) - A(\theta) + C(Y, Z),$$

where $S(Y, Z)$ is the complete-data sufficient statistic, $\eta(\theta)$ the natural parameter, and $A(\theta)$ the log-partition function. By linearity of expectation,

$$Q(\theta \mid \theta^{(t)}) = \eta(\theta)^\top \tilde{S}^{(t)} - A(\theta) + \text{const}, \quad \tilde{S}^{(t)} := \mathbb{E}_{\theta^{(t)}}\{S(Y, Z) \mid Y\}.$$

Key Idea

In exponential-family problems the E-step reduces to computing the expected complete-data sufficient statistic $\tilde{S}^{(t)}$; the M-step then maximizes as if $\tilde{S}^{(t)}$ were the actually-observed statistic – i.e., plug $\tilde{S}^{(t)}$ into whatever formula gives the ordinary MLE if the complete data had really been observed. In the mixture problem, S collects $n_k = \sum_i Z_{ik}, \sum_i Z_{ik}y_i, \sum_i Z_{ik}y_i^2$; the E-step replaces Z_{ik} by $\tau_{ik}^{(t)}$ throughout. In the incomplete-table problems, S is the cell counts, and the E-step computes their conditional expectations.

A general 5-step EM recipe

1. Choose the complete data $X = (Y, Z)$.
2. Write the complete-data log-likelihood $\ell_c(\theta; Y, Z)$.
3. Identify the complete-data sufficient statistics $S(Y, Z)$.
4. **E-step** – compute $\tilde{S}^{(t)} = \mathbb{E}_{\theta^{(t)}}\{S(Y, Z) \mid Y\}$.
5. **M-step** – maximize as if $\tilde{S}^{(t)}$ were observed: $\theta^{(t+1)} = \arg \max_{\theta} Q(\theta \mid \theta^{(t)})$.

Problem	Missing part	E-step computes
Normal mixture	component labels	responsibilities τ_{ik}
Incomplete multinomial	missing cell split	expected cell counts
Incomplete contingency table	missing row/column	expected cell counts
Bivariate normal (missing coord.)	missing coordinate	conditional moments (mean <i>and</i> variance)

8.1 Full derivations (Lagrangian and calculus)

Mixing proportions π_k . With $n_k = \sum_i Z_{ik}$ (or $\sum_i \tau_{ik}^{(t)}$ for the EM update), the relevant term is $\ell_{c,\pi} = \sum_k n_k \log \pi_k$ subject to $\sum_k \pi_k = 1$. The Lagrangian $L = \sum_k n_k \log \pi_k + \lambda(\sum_k \pi_k - 1)$ gives $\partial L / \partial \pi_k = n_k / \pi_k + \lambda = 0 \Rightarrow \pi_k = -n_k / \lambda$; summing, $1 = -\frac{1}{\lambda} \sum_k n_k = -n / \lambda$, so $\lambda = -n$ and $\hat{\pi}_k = n_k / n$.

Component means μ_k . Terms involving μ_k : $-\frac{1}{2\sigma_k^2} \sum_i Z_{ik}(y_i - \mu_k)^2$. Minimizing $S_k(\mu_k) = \sum_i Z_{ik}(y_i - \mu_k)^2$: $dS_k/d\mu_k = -2 \sum_i Z_{ik}(y_i - \mu_k) = 0 \Rightarrow \hat{\mu}_k = \sum_i Z_{ik}y_i / \sum_i Z_{ik}$.

Component variances σ_k^2 . Let $v_k = \sigma_k^2$, $A_k = \sum_i Z_{ik}(y_i - \hat{\mu}_k)^2$. The relevant term is $g(v_k) = -\frac{n_k}{2} \log v_k - \frac{A_k}{2v_k}$; $g'(v_k) = -\frac{n_k}{2v_k} + \frac{A_k}{2v_k^2} = 0 \Rightarrow -n_k v_k + A_k = 0 \Rightarrow \hat{\sigma}_k^2 = A_k/n_k = \sum_i Z_{ik}(y_i - \hat{\mu}_k)^2 / \sum_i Z_{ik}$.

Exam tip

All M-step derivations above (mixture π_k, μ_k, σ_k^2 ; multinomial p_k ; contingency-table p_{ij}) follow the same two techniques: a Lagrange multiplier for constrained proportions summing to 1, and ordinary calculus (derivative = 0) for unconstrained means/variances. Exams frequently ask you to reproduce these derivations for a structurally similar but new scenario.

8.2 Standard errors after EM

EM produces point estimates but not, by itself, standard errors – Q is a device for optimization, not a likelihood whose curvature directly gives the observed information. Two standard remedies:

- **Louis's method (1982).** The observed information decomposes as $I_{\text{obs}}(\hat{\theta}) = I_{\text{complete}}(\hat{\theta}) - I_{\text{missing}}(\hat{\theta})$, where $I_{\text{complete}}(\hat{\theta}) = -\mathbb{E}_{\hat{\theta}}[\partial^2 \ell_c / \partial \theta \partial \theta^\top | Y]$ evaluated at $\hat{\theta}$, and $I_{\text{missing}}(\hat{\theta})$ is the conditional variance of the complete-data score given Y – both computable from quantities already produced during the E-step.
- **Bootstrap (Efron & Tibshirani, 1994).** Resample the observed data (or simulate from the fitted model), re-run EM to convergence on each resample, and use the empirical variability of $\hat{\theta}^{(b)}$ across resamples to estimate standard errors – computationally simple but more expensive.

9 Quick-Reference Formula Sheet

Normal mixture

$$\tau_{ik}^{(t)} = \frac{\pi_k^{(t)} \phi(y_i; \mu_k^{(t)}, \sigma_k^{2(t)})}{\sum_\ell \pi_\ell^{(t)} \phi(y_i; \mu_\ell^{(t)}, \sigma_\ell^{2(t)})},$$

$$\pi_k^{(t+1)} = \frac{1}{n} \sum_i \tau_{ik}^{(t)}, \quad \mu_k^{(t+1)} = \frac{\sum_i \tau_{ik}^{(t)} y_i}{\sum_i \tau_{ik}^{(t)}}, \quad \sigma_k^{2(t+1)} = \frac{\sum_i \tau_{ik}^{(t)} (y_i - \mu_k^{(t+1)})^2}{\sum_i \tau_{ik}^{(t)}}.$$

Incomplete two-way table

$$\tilde{n}_{ij}^{(t)} = c_{ij} + a_i \frac{p_{ij}^{(t)}}{p_{i+}} + b_j \frac{p_{ij}^{(t)}}{p_{+j}} + r p_{ij}^{(t)}, \quad p_{ij}^{(t+1)} = \tilde{n}_{ij}^{(t)} / N.$$

Conditional bivariate normal & exponential-family Q

$$Y | X = x \sim N\left(\mu_y + \frac{\sigma_{xy}}{\sigma_{xx}}(x - \mu_x), \sigma_{yy} - \frac{\sigma_{xy}^2}{\sigma_{xx}}\right), \quad Q(\theta | \theta^{(t)}) = \eta(\theta)^\top \tilde{S}^{(t)} - A(\theta) + c.$$

Common exam pitfalls – final checklist

- Confusing τ_{ik} (a posterior probability in $[0, 1]$) with a hard 0/1 assignment: EM performs *soft* clustering.
- Forgetting constraints ($\sum_k \pi_k = 1$, $\sum p_{ij} = 1$) when deriving an M-step – always introduce a Lagrange multiplier.
- Using the old mean $\mu_k^{(t)}$ instead of the updated mean $\mu_k^{(t+1)}$ in the variance update.

- For overlapping ambiguous groups, forgetting a category can receive expected contributions from *multiple* ambiguous groups simultaneously.
- Forgetting the conditional *second moment* (not just the conditional mean) when imputing missing continuous values for variance/covariance updates.
- Overstating the ascent property: EM never decreases the observed log-likelihood, but does not guarantee convergence to the global maximum, especially for multimodal likelihoods.

Part II

The MM Algorithm

10 Motivation: Beyond EM

EM handles missing/latent-data likelihoods via conditional expectation. More broadly, many useful iterative algorithms work by replacing a hard objective with an easier **surrogate** objective at each iteration – the central idea of **MM algorithms** (Hunter and Lange, 2004; Lange, 2016). We wish to solve $\hat{\theta} = \arg \max_{\theta} \ell(\theta)$ or $\hat{\theta} = \arg \min_{\theta} f(\theta)$, where the objective may be non-quadratic, contain sums inside logarithms, be nondifferentiable, or admit no closed-form update.

Goal	Surrogate relation	Update
Minimize $f(\theta)$	majorize f by $g(\theta \mid \theta^{(t)})$	minimize g
Maximize $\ell(\theta)$	minorize ℓ by $q(\theta \mid \theta^{(t)})$	maximize q

Key Idea

MM means: do not attack the mountain directly. Build a local, easy-to-optimize approximation whose improvement forces improvement in the original objective. The two letters are used in two directions: *majorization-minimization* for minimizing a loss, and *minorization-maximization* for maximizing a likelihood – same logic, opposite direction, sometimes called *optimization transfer* (Lange, Hunter, and Yang, 2000).

11 Majorization-Minimization

Definition 11.1 (Majorization). $g(\theta \mid \theta^{(t)})$ majorizes $f(\theta)$ at $\theta^{(t)}$ if $g(\theta \mid \theta^{(t)}) \geq f(\theta)$ for all θ (*domination*) and $g(\theta^{(t)} \mid \theta^{(t)}) = f(\theta^{(t)})$ (*tangency*).

Given a majorizer, set $\theta^{(t+1)} \in \arg \min_{\theta} g(\theta \mid \theta^{(t)})$.

Theorem 11.2 (Descent property). If $g(\theta \mid \theta^{(t)})$ majorizes f at $\theta^{(t)}$ and $\theta^{(t+1)} \in \arg \min_{\theta} g(\theta \mid \theta^{(t)})$, then $f(\theta^{(t+1)}) \leq f(\theta^{(t)})$.

Proof. (1) By majorization, $f(\theta^{(t+1)}) \leq g(\theta^{(t+1)} \mid \theta^{(t)})$. (2) Since $\theta^{(t+1)}$ minimizes g , $g(\theta^{(t+1)} \mid \theta^{(t)}) \leq g(\theta^{(t)} \mid \theta^{(t)})$. (3) By tangency, $g(\theta^{(t)} \mid \theta^{(t)}) = f(\theta^{(t)})$. Chaining the three gives $f(\theta^{(t+1)}) \leq f(\theta^{(t)})$. ■ □

Key Idea

Geometrically: the surrogate surface sits above the true objective everywhere and touches it at the current point; we move to where the surrogate is smaller; since the true objective lies below the surrogate everywhere, it cannot have increased. The surrogate may be crude away from $\theta^{(t)}$, but it must be tight at $\theta^{(t)}$ and bound the objective in the correct direction.

12 Minorization-Maximization

Definition 12.1 (Minorization). $q(\theta \mid \theta^{(t)})$ minorizes $\ell(\theta)$ at $\theta^{(t)}$ if $q(\theta \mid \theta^{(t)}) \leq \ell(\theta)$ for all θ , with $q(\theta^{(t)} \mid \theta^{(t)}) = \ell(\theta^{(t)})$.

Given a minorizer, set $\theta^{(t+1)} \in \arg \max_{\theta} q(\theta \mid \theta^{(t)})$.

Theorem 12.2 (Ascent property). If $q(\theta \mid \theta^{(t)})$ minorizes ℓ at $\theta^{(t)}$ and $\theta^{(t+1)} \in \arg \max_{\theta} q(\theta \mid \theta^{(t)})$, then $\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})$.

Proof. Mirrors Theorem 11.2 with inequalities reversed: $\ell(\theta^{(t+1)}) \geq q(\theta^{(t+1)} | \theta^{(t)}) \geq q(\theta^{(t)} | \theta^{(t)}) = \ell(\theta^{(t)})$. $\blacksquare$ $\square$

Remark 12.3. Exact maximization of the surrogate is not required for the ascent property to hold: if $q(\theta^{(t+1)} | \theta^{(t)}) \geq q(\theta^{(t)} | \theta^{(t)})$ (a *generalized* MM step that merely improves q rather than maximizing it), the same three-line chain still gives $\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})$. This is practically important: MM remains a valid ascent method even when the surrogate subproblem is solved only approximately.

13 Constructing Surrogate Functions

Common tools for building majorizers/minorizers:

13.1 Tangent-line inequalities

If h is concave, $h(u) \leq h(u_0) + h'(u_0)(u - u_0)$ (the tangent line lies above); if h is convex, $h(u) \geq h(u_0) + h'(u_0)(u - u_0)$ (the tangent line lies below). These convert nonlinear terms into linear or quadratic surrogate terms.

13.2 Quadratic (Lipschitz-gradient) bounds

If f is differentiable with L -Lipschitz gradient ($\|\nabla f(\theta) - \nabla f(\eta)\| \leq L\|\theta - \eta\|$), then $f(\theta) \leq f(\theta^{(t)}) + \nabla f(\theta^{(t)})^\top (\theta - \theta^{(t)}) + \frac{L}{2}\|\theta - \theta^{(t)}\|^2$ is a quadratic majorizer. Minimizing it: differentiate and set to zero, $\nabla f(\theta^{(t)}) + L(\theta - \theta^{(t)}) = 0 \Rightarrow \theta^{(t+1)} = \theta^{(t)} - \frac{1}{L}\nabla f(\theta^{(t)})$ – gradient descent with step size $1/L$, understood as an MM algorithm.

13.3 Jensen's inequality

For weights $w_k > 0$, $\sum_k w_k = 1$,

$$\log\left(\sum_k a_k\right) = \log\left(\sum_k w_k \frac{a_k}{w_k}\right) \geq \sum_k w_k \log\left(\frac{a_k}{w_k}\right),$$

by concavity of $\log$, with equality when $w_k \propto a_k$. For a mixture log-likelihood $\ell(\theta) = \sum_i \log\{\sum_k \pi_k f_k(y_i; \psi_k)\}$, introducing weights $w_{ik} \geq 0$, $\sum_k w_{ik} = 1$ gives $\log\{\sum_k \pi_k f_k(y_i; \psi_k)\} \geq \sum_k w_{ik} \log\{\pi_k f_k(y_i; \psi_k)/w_{ik}\}$, with equality at $\theta^{(t)}$ achieved by $w_{ik} = \tau_{ik}^{(t)}$ – exactly the EM responsibilities. **This is the key inequality behind the EM lower bound for mixture likelihoods**, and shows EM is a special MM algorithm.

14 EM as an MM Algorithm

Recall $L(\theta; Y_{\text{obs}}) = \int L_c(\theta; Y_{\text{obs}}, y_{\text{mis}}) dy_{\text{mis}}$ and $\ell(\theta) = \log L(\theta; Y_{\text{obs}})$. Defining $Q(\theta | \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}}[\ell_c(\theta; Y_{\text{com}} | Y_{\text{obs}})]$, the decomposition $\ell(\theta) = Q(\theta | \theta^{(t)}) + [\text{a term not depending on } \theta, \text{ maximized at } \theta = \theta^{(t)}]$ (exactly $-H(\theta | \theta^{(t)})$ from Theorem 2.1's proof) shows Q is a genuine minorizer of ℓ touching it at $\theta^{(t)}$. Maximizing Q therefore guarantees ascent in ℓ by the general MM theorem – EM's ascent property is a corollary of the MM ascent property.

Feature	EM	General MM
Surrogate source	conditional expectation	any valid inequality
Missing data needed?	yes, conceptually	no
Guarantee	likelihood ascent	ascent/descent
Update style	E-step / M-step	construct / optimize surrogate

15 Worked Example I: Robust Regression (LAD)

Least Absolute Deviations regression, $\hat{\beta}_{\text{LAD}} = \arg \min_{\beta} \sum_i |y_i - x_i^\top \beta|$, is useful because absolute loss is less sensitive to outliers than squared loss, but is nondifferentiable at zero residuals.

Lemma 15.1. For any $u \in \mathbb{R}$ and $u^{(t)} \neq 0$: $|u| \leq \frac{u^2}{2|u^{(t)}|} + \frac{|u^{(t)}|}{2}$, with equality at $u = u^{(t)}$.

Proof. Since squares are nonnegative, $0 \leq (|u| - |u^{(t)}|)^2 = u^2 - 2|u||u^{(t)}| + |u^{(t)}|^2$, so $2|u||u^{(t)}| \leq u^2 + |u^{(t)}|^2$, giving the claim after dividing by $2|u^{(t)}|$. Equality holds at $|u| = |u^{(t)}|$. $\square$

Let $r_i(\beta) = y_i - x_i^\top \beta$, $r_i^{(t)} = r_i(\beta^{(t)})$. Applying Lemma 15.1 termwise,

$$g(\beta \mid \beta^{(t)}) = \sum_{i=1}^n \left[\frac{(y_i - x_i^\top \beta)^2}{2|r_i^{(t)}|} + \frac{|r_i^{(t)}|}{2} \right]$$

majorizes the LAD objective. Dropping the β -free constant, minimizing g is a weighted least squares problem $\sum_i w_i^{(t)} (y_i - x_i^\top \beta)^2$ with $w_i^{(t)} = 1/(2|r_i^{(t)}|)$, solved in closed form:

$$\beta^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} y, \quad W^{(t)} = \text{diag}(w_1^{(t)}, \dots, w_n^{(t)}).$$

Numerical safeguard

Replace $|r_i^{(t)}|$ by $\sqrt{(r_i^{(t)})^2 + \varepsilon}$ or $|r_i^{(t)}| + \varepsilon$ to avoid division by zero when a residual is (near) zero. This is precisely **Iteratively Reweighted Least Squares (IRLS)**, derived entirely from the MM principle.

16 Worked Example II: A Scalar Penalty Problem

Consider $f(\theta) = \frac{1}{2}(y - \theta)^2 + \lambda|\theta|$, $\lambda > 0$ – a scalar LASSO-type problem, nondifferentiable at $\theta = 0$. Using Lemma 15.1,

$$g(\theta \mid \theta^{(t)}) = \frac{1}{2}(y - \theta)^2 + \lambda \left(\frac{\theta^2}{2|\theta^{(t)}|} + \frac{|\theta^{(t)}|}{2} \right).$$

Dropping constants and differentiating $g(\theta \mid \theta^{(t)}) = \frac{1}{2}(y - \theta)^2 + \frac{\lambda}{2|\theta^{(t)}|}\theta^2$: $\partial g / \partial \theta = -(y - \theta) + \frac{\lambda}{|\theta^{(t)}|}\theta = 0 \Rightarrow \theta(1 + \frac{\lambda}{|\theta^{(t)}|}) = y$, giving

$$\theta^{(t+1)} = \frac{|\theta^{(t)}|}{|\theta^{(t)}| + \lambda} y.$$

This shrinks y toward zero more strongly as λ grows or as $|\theta^{(t)}|$ shrinks.

Comparison with the exact solution

The scalar LASSO problem actually has a closed-form *soft-thresholding* solution, $\hat{\theta} = \text{sign}(y)(|y| - \lambda)_+$, so MM is not the most efficient route here – but the example cleanly shows how MM turns nonsmooth optimization into a sequence of smooth (quadratic) subproblems, which *does* generalize to vector-valued LASSO/penalized problems without closed-form solutions.

17 MM for Likelihood and GLM-Type Problems

For a concave log-likelihood $\ell(\beta)$ that is not quadratic, a quadratic minorizer is

$$q(\beta \mid \beta^{(t)}) = \ell(\beta^{(t)}) + \nabla \ell(\beta^{(t)})^\top (\beta - \beta^{(t)}) - \frac{M}{2} \|\beta - \beta^{(t)}\|^2,$$

for M large enough to bound the curvature of ℓ from below. Maximizing: $\nabla_{\beta} q = \nabla \ell(\beta^{(t)}) - M(\beta - \beta^{(t)}) = 0 \Rightarrow \beta^{(t+1)} = \beta^{(t)} + \frac{1}{M} \nabla \ell(\beta^{(t)})$ – gradient ascent as minorization-maximization.

17.1 Comparison with Newton's method

Newton's method, $\theta^{(t+1)} = \theta^{(t)} - \{\nabla^2 f(\theta^{(t)})\}^{-1} \nabla f(\theta^{(t)})$, uses local curvature and converges very fast near the optimum but may fail to decrease f globally without safeguards (line search, trust regions). MM instead prioritizes monotonicity at the cost of potentially slower convergence if the surrogate is conservative.

Method	Core idea	Strength
EM	conditional expectation	natural for missing data
MM	surrogate optimization	very general
Newton	local quadratic approximation	fast near optimum
Gradient descent	first-order quadratic bound	simple, scalable

18 Choosing Good Surrogates

A good surrogate must be (1) *computationally simple* – easier to optimize than the original objective, and (2) *tight* – not so loose that it forces overly cautious, slow-converging steps. Common stopping criteria: small parameter change $\|\theta^{(t+1)} - \theta^{(t)}\| < \varepsilon$; small relative objective change $|f(\theta^{(t+1)}) - f(\theta^{(t)})|/(1 + |f(\theta^{(t)})|) < \varepsilon$; small gradient/score norm; or a maximum iteration count. As with EM, monotone improvement does not by itself certify a global optimum – MM inherits the usual difficulties of nonconvex optimization (multiple stationary points, starting-value sensitivity).

19 Exercises

Verify a majorizer

Let $f(\theta) = |\theta|$. For $\theta^{(t)} \neq 0$, show $g(\theta | \theta^{(t)}) = \frac{\theta^2}{2|\theta^{(t)}|} + \frac{|\theta^{(t)}|}{2}$ majorizes f at $\theta^{(t)}$, then find the MM update for minimizing $\frac{1}{2}(y - \theta)^2 + \lambda|\theta|$.

Solution. This is exactly Lemma 15.1 and Section 14 above; the update is $\theta^{(t+1)} = \frac{|\theta^{(t)}|}{|\theta^{(t)}| + \lambda} y$.

Ascent without exact maximization

Suppose $q(\theta | \theta^{(t)})$ minorizes ℓ at $\theta^{(t)}$, and $\theta^{(t+1)}$ only *improves* q (i.e. $q(\theta^{(t+1)} | \theta^{(t)}) \geq q(\theta^{(t)} | \theta^{(t)})$) rather than maximizing it exactly. Does $\ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)})$ still hold?

Solution. Yes: $\ell(\theta^{(t+1)}) \geq q(\theta^{(t+1)} | \theta^{(t)}) \geq q(\theta^{(t)} | \theta^{(t)}) = \ell(\theta^{(t)})$ by minorization, the assumed improvement, and tangency respectively. This is sometimes called a *generalized MM step*, and remains valid even if the surrogate subproblem is only approximately solved.

The MM algorithm: full recipe (minimization form)

1. Start with $\theta^{(0)}$.
2. At iteration t , construct $g(\theta | \theta^{(t)})$ with $g \geq f$ everywhere and $g(\theta^{(t)} | \theta^{(t)}) = f(\theta^{(t)})$.
3. Set $\theta^{(t+1)} = \arg \min_{\theta} g(\theta | \theta^{(t)})$.
4. Repeat until convergence. (For maximization, reverse the inequalities and maximize the minorizer.)

Chapter summary

Bound the difficult objective, optimize the easier bound, and inherit monotone improvement. EM is a special and important MM-type algorithm for incomplete-data likelihoods; MM more broadly handles

robust regression, penalized estimation, GLMs, and many other optimization problems.

Part III

Survival Analysis

20 Time-to-Event Data, Survival, and Hazard

20.1 Why survival analysis is different

Lectures I–VII repeatedly showed that the data we wish we had and the data we observe may differ: missing covariates led to missing-data mechanisms; latent labels led to EM for mixtures; incomplete cell counts led to EM for tables; difficult objectives led to MM. Time-to-event data with **censoring** is another major kind of incomplete observation, requiring its own likelihood-based framework (Kleinbaum and Klein, 2012; Kalbfleisch and Prentice, 2002).

Let $T \geq 0$ be a time-to-event variable – the time from a well-defined *origin* to a well-defined *event*, on a specified *time scale*. If some individuals have not yet experienced the event when the study ends, replacing their event time by the last follow-up time would be wrong: the last follow-up time is not the true event time – it only tells us the true time is larger.

Example 20.1 (Motivating dataset).	Patient	Follow-up time	Status δ_i	Meaning
	1	4	1	relapse observed at time 4
	2	6	0	relapse-free until time 6
	3	7	1	relapse observed at time 7
	4	9	0	relapse-free until time 9
	5	10	1	relapse observed at time 10

For patient 2, $(U_2, \delta_2) = (6, 0)$ does *not* mean $T_2 = 6$; it means $T_2 > 6$. A censored observation is incomplete but still informative – an inequality rather than an exact value.

20.2 Survival, density, and hazard

For $T \geq 0$: distribution function $F(t) = P(T \leq t)$; **survival function** $S(t) = P(T > t) = 1 - F(t)$ – the probability of remaining event-free beyond t , nonincreasing, with $S(0) \leq 1$ and $S(t) \rightarrow 0$ as $t \rightarrow \infty$. If T is continuous with density $f(t) = F'(t) = -S'(t)$.

Definition 20.2 (Hazard function).

$$h(t) = \lim_{\Delta t \downarrow 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}.$$

Proposition 20.3. For continuous T , $h(t) = f(t)/S(t)$.

Proof. $P(t \leq T < t + \Delta t \mid T \geq t) = P(t \leq T < t + \Delta t)/P(T \geq t) \approx f(t)\Delta t/S(t)$ for small Δt (using $P(T \geq t) = P(T > t) = S(t)$ for continuous T). Dividing by Δt and letting $\Delta t \downarrow 0$ gives $h(t) = f(t)/S(t)$. $\square$

Caution

A hazard is *not* a probability – it is a rate and may exceed 1 depending on the time scale. $h(12) = 0.04/\text{month}$ means that near month 12, *among those still event-free*, the instantaneous event rate is about 0.04 per month; it does not mean a 0.04 probability over the whole study. The phrase “among those still at risk” is essential – hazards are conditional on survival up to that time.

Definition 20.4 (Cumulative hazard). $H(t) = \int_0^t h(u) du$.

Proposition 20.5. $H(t) = -\log S(t)$, equivalently $S(t) = \exp\{-H(t)\}$.

Proof. $h(t) = f(t)/S(t) = -S'(t)/S(t) = -\frac{d}{dt} \log S(t)$. Integrating from 0 to t with $S(0) = 1$: $H(t) = \int_0^t h(u) du = -[\log S(u)]_0^t = -\log S(t)$. $\square$

Quantity	Question answered
$S(t)$	What proportion remains event-free beyond t ?
$f(t)$	Around which times do events tend to occur?
$h(t)$	Among those at risk at t , how fast are events occurring now?
$H(t)$	How much event risk has accumulated up to t ?

Example 20.6 (Exponential and Weibull survival). If $T \sim \text{Exponential}(\lambda)$: $f(t) = \lambda e^{-\lambda t}$, $S(t) = e^{-\lambda t}$, so $h(t) = \lambda$ – **constant hazard**, a useful but often unrealistic starting model. For the Weibull parametrization $S(t) = \exp\{-\lambda t^\alpha\}$: $H(t) = \lambda t^\alpha$, so $h(t) = H'(t) = \lambda \alpha t^{\alpha-1}$. The shape α controls hazard behavior: $\alpha = 1$ constant, $\alpha > 1$ increasing (aging/wear-out), $0 < \alpha < 1$ decreasing (early fragile-unit failures).

Quick check

Suppose $S(t) = \exp(-2t^3)$. Find $H(t), h(t), f(t)$.

Solution. $H(t) = -\log S(t) = 2t^3$; $h(t) = H'(t) = 6t^2$; $f(t) = h(t)S(t) = 6t^2 \exp(-2t^3)$.

21 Censoring

21.1 Types of censoring

Censoring occurs when T is not observed exactly but partial information is available – a set of possible values rather than one exact value.

- **Right censoring:** $T > C$. We observe $U = \min(T, C)$, $\delta = \mathbf{1}(T \leq C)$; $\delta = 0 \Rightarrow U = C < T$. Sources: administrative end of study, loss to follow-up, dropout, or “no event by analysis date.”
- **Left censoring:** $T \leq C$ – we only know the event occurred before a certain time (e.g., an infection detected at first positive test, exact infection time unknown).
- **Interval censoring:** $L < T \leq R$ – the event time lies in an interval (e.g., periodic medical screening: negative at month 6, positive at month 12).

Censoring vs. truncation

Censoring: the subject is in the sample, but the exact event time is not fully observed. **Truncation:** the subject is not included in the sample unless their event time satisfies a condition (e.g., a retrospective study enrolling only those who survived at least 5 years – *left truncation*). Truncation needs different likelihood adjustments and should not be treated as ordinary censoring.

Classify the censoring/truncation type

(1) Event-free at the final visit at month 24. (2) Disease already present at the first diagnostic test. (3) Disease-free in January, diseased in April. (4) Only individuals surviving ≥ 5 years post-diagnosis are included retrospectively. (5) A machine is still working when the experiment ends.

Solution. (1) Right censored. (2) Left censored. (3) Interval censored. (4) Left truncated. (5) Right censored.

Rule of thumb: censoring is partial information for a sampled unit; truncation affects whether the unit enters the sample at all.

21.2 Independent censoring

Standard survival methods assume censoring is **independent** (noninformative): with covariates X , $T \perp C \mid X$ (reducing to $T \perp C$ with no covariates) – after conditioning on covariates, the censoring mechanism carries no extra information about the event time.

Plausibly independent	Possibly informative
Study ends on a fixed administrative date	Patients drop out because symptoms worsen
Device removed due to unrelated relocation	Machine removed because visibly deteriorating
Follow-up stops due to funding end date	High-risk participants become harder to contact

Connection to missing-data mechanisms

Censoring is a structured cousin of the MAR/MCAR framework of Section 1: exact event time not observed $\leftrightarrow$ missing value; censoring mechanism $\leftrightarrow$ missingness mechanism; independent censoring $\leftrightarrow$ ignorable missingness. But censoring is *not* MCAR: $\delta_i = 0 \Rightarrow T_i > U_i$, so the censoring status directly constrains the unobserved event time. Deleting censored observations (a “complete-case” approach) is not merely inefficient here – it systematically biases estimation, typically underestimating survival, since the individuals most likely to survive longest are disproportionately censored.

22 Likelihood Contributions and Parametric Models

22.1 Likelihood by observation type

The likelihood contribution must match the information actually observed:

Observed information	Meaning	Likelihood contribution
Exact event at t	$T = t$	$f(t; \theta)$
Right censored at c	$T > c$	$S(c; \theta)$
Left censored at c	$T \leq c$	$F(c; \theta)$
Interval censored in $(a, b]$	$a < T \leq b$	$S(a; \theta) - S(b; \theta)$

22.2 The right-censored likelihood

For (U_i, δ_i) , $U_i = \min(T_i, C_i)$, $\delta_i = \mathbf{1}(T_i \leq C_i)$, the per-subject contribution is $L_i(\theta) = \{f(u_i; \theta)\}^{\delta_i} \{S(u_i; \theta)\}^{1-\delta_i}$, and for n independent subjects,

$$L(\theta) = \prod_{i=1}^n \{f(u_i; \theta)\}^{\delta_i} \{S(u_i; \theta)\}^{1-\delta_i}.$$

Proposition 22.1. *Equivalently, $L(\theta) = \prod_i \{h(u_i; \theta)\}^{\delta_i} S(u_i; \theta)$.*

Proof. Since $f = hS$,

$$\begin{aligned} \{f(u_i; \theta)\}^{\delta_i} \{S(u_i; \theta)\}^{1-\delta_i} &= \{h(u_i; \theta)S(u_i; \theta)\}^{\delta_i} \{S(u_i; \theta)\}^{1-\delta_i} \\ &= \{h(u_i; \theta)\}^{\delta_i} S(u_i; \theta)^{\delta_i + 1 - \delta_i} = \{h(u_i; \theta)\}^{\delta_i} S(u_i; \theta). \end{aligned} \quad \square$$

Taking logs, $\ell(\theta) = \sum_i [\delta_i \log h(u_i; \theta) + \log S(u_i; \theta)]$. This hazard form cleanly separates: *everyone* contributes $\log S(u_i; \theta)$ (survival up to their own observed time), while only uncensored subjects *additionally* contribute $\delta_i \log h(u_i; \theta)$ (the event occurring exactly then).

Do (8, 1) and (8, 0) mean the same thing?

No. (8, 1): event at exactly $t = 8$, contribution $f(8)$. (8, 0): event-free through 8, $T > 8$, contribution $S(8)$. The same recorded time carries different information depending on δ .

22.3 The exponential model

$f(t; \lambda) = \lambda e^{-\lambda t}$, $S(t; \lambda) = e^{-\lambda t}$, so $L(\lambda) = \prod_i \{\lambda e^{-\lambda u_i}\}^{\delta_i} \{e^{-\lambda u_i}\}^{1-\delta_i} = \lambda^d \exp\{-\lambda \sum_i u_i\}$, where $d = \sum_i \delta_i$ is the number of observed events. Then $\ell(\lambda) = d \log \lambda - \lambda \sum_i u_i$, and

$$\frac{d\ell}{d\lambda} = \frac{d}{\lambda} - \sum_i u_i = 0 \implies \hat{\lambda} = \frac{d}{\sum_i u_i}, \quad \frac{d^2\ell}{d\lambda^2} = -\frac{d}{\lambda^2} < 0 \text{ (if } d > 0\text{),}$$

confirming a maximum. The denominator $\sum_i u_i$ is the total observed *person-time*, including censored subjects' follow-up – censored subjects contribute time-at-risk even though they contribute no event.

Example 22.2 (Worked: administrative censoring). Data $(u_i) = (3, 5, 6, 8, 11, 12, 15, 18)$, $(\delta_i) = (1, 0, 1, 1, 0, 1, 0, 1)$: $d = 5$, $\sum_i u_i = 3 + 5 + 6 + 8 + 11 + 12 + 15 + 18 = 78$, so $\hat{\lambda} = 5/78 = 0.0641$.

22.4 The Weibull model

$S(t; k, \eta) = \exp\{-(t/\eta)^k\}$, $f(t; k, \eta) = \frac{k}{\eta}(t/\eta)^{k-1} \exp\{-(t/\eta)^k\}$, $h(t; k, \eta) = \frac{k}{\eta}(t/\eta)^{k-1}$, with shape k (constant/increasing/decreasing hazard as $k = 1, > 1, < 1$) and scale η . Using $\log h(u_i; k, \eta) = \log k - \log \eta + (k-1) \log(u_i/\eta)$ and $\log S(u_i; k, \eta) = -(u_i/\eta)^k$,

$$\ell(k, \eta) = d \log k - dk \log \eta + (k-1) \sum_i \delta_i \log u_i - \sum_i \left(\frac{u_i}{\eta}\right)^k.$$

For fixed k , $\partial \ell / \partial \eta = -\frac{dk}{\eta} + \frac{k}{\eta^{k+1}} \sum_i u_i^k = 0 \implies \hat{\eta}(k)^k = \sum_i u_i^k / d$: η has a closed-form update given k , but k itself generally needs a numerical (profile-likelihood) search – reducing a 2-D optimization to a 1-D search over k . For the administrative-censoring data above, the profile maximum occurs near $\hat{k} \approx 1.99$, $\hat{\eta} \approx 13.77$.

22.5 The Gamma model and general parametric summary

$T \sim \text{Gamma}(\alpha, \beta)$: $f(t; \alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t}$; unlike the exponential, $S(t; \alpha, \beta)$ involves the incomplete gamma function with no simple closed form, so both the survival function and the MLE typically require numerical routines. For the same administrative-censoring dataset, a grid search gives $\hat{\alpha} \approx 2.78$, $\hat{\beta} \approx 0.218$.

Model	Typical use	Hazard	MLE with censoring
Exponential	first simple model	constant	often analytic
Weibull	reliability, medical	monotone	partly analytic, partly numerical
Gamma	flexible waiting time	non-monotone	numerical
Log-normal	skewed event times	non-monotone	numerical
Log-logistic	long-tailed survival	non-monotone	numerical

22.6 The general MLE problem and why censoring is not MCAR

For general θ , $\hat{\theta} = \arg \max_\theta \ell(\theta)$ with score $U(\theta) = \sum_i [\delta_i \partial_\theta \log f(u_i; \theta) + (1 - \delta_i) \partial_\theta \log S(u_i; \theta)]$. Censored observations are *not* absent from the score equation – they contribute through $\partial_\theta \log S(u_i; \theta)$, so censored observations still carry information about θ . As emphasized above, $\delta_i = 0 \implies T_i > U_i$ directly constrains the unobserved event time; this is why naive complete-case deletion of censored records is invalid, unlike genuinely MCAR missingness.

23 Nonparametric Estimation of the Survival Curve

23.1 Motivation and risk sets

Rather than assuming exponential/Weibull/gamma, we can estimate $S(t)$ *nonparametrically*. With complete data, the natural empirical survival curve is $\hat{S}_{\text{emp}}(t) = \frac{1}{n} \sum_i \mathbf{1}(t_i > t)$, a right-continuous step function

dropping by $1/n$ at each failure. With censoring, the challenge is deciding who remains “in the denominator” at each event time.

For distinct uncensored event times $y_{(1)} < \dots < y_{(k)}$, define the **risk set** $R_j = \{i : Y_i \geq y_{(j)}\}$, its size $N_j = |R_j|$ (individuals still under observation and event-free just before $y_{(j)}$), and d_j = number of events exactly at $y_{(j)}$.

23.2 The product-limit idea and the Kaplan–Meier estimator

Since $P(T > t_3) = P(T > t_1)P(T > t_2 \mid T > t_1)P(T > t_3 \mid T > t_2)$, we estimate each local conditional survival probability by $1 - d_j/N_j$ and multiply:

Definition 23.1 (Kaplan–Meier / product-limit estimator).

$$\hat{S}(t) = \prod_{j: y_{(j)} \leq t} \left(1 - \frac{d_j}{N_j}\right).$$

Example 23.2 (Worked two-cohort study). 100 patients recruited in year 0, followed 2 years; 1000 patients recruited in year 1, followed 1 year. Deaths: 70 (cohort 1, year 1), 15 (cohort 1, year 2), 750 (cohort 2, year 1). At $t = 1$: of 1100 total at risk, $30 + 250 = 280$ survive year 1, so $\hat{P}(T > 1) = 280/1100$. Only cohort 1’s 30 survivors remain at risk in year 2, of whom 15 die: $\hat{P}(T > 2 \mid T > 1) = 15/30$. Hence $\hat{S}(2) = \frac{280}{1100} \times \frac{15}{30} = 0.127$. Naive alternatives (reduced sample, assume censored died, assume censored survived, delete censored) give badly biased answers of 0.150, 0.014, 0.241, 0.018 respectively – illustrating why ad hoc handling of censoring is unreliable.

Example 23.3 (Small worked example with ties). Data 3, 2⁺, 0, 1, 5⁺, 3, 5 (+ = censored). Uncensored event times: 0, 1, 3, 5.

$y_{(j)}$	N_j	d_j	factor $1 - d_j/N_j$	$\hat{S}(y_{(j)})$
0	7	1	6/7	0.86
1	6	1	5/6	0.71
3	4	2	1/2	0.36
5	2	1	1/2	0.18

At $t = 3$ there are two failures ($d_j = 2$), giving a larger single drop of factor $1/2$; censoring marks (at 2 and 5) do not themselves create a drop, though censoring at 2 shrinks the risk set going into $t = 3$.

Handling ties

Count all failures at a tied time in d_j and apply one multiplicative factor $1 - d_j/N_j$; censored observations recorded at the same time as failures are conventionally treated as occurring just after the failures, unless a study-specific convention differs.

23.3 Kaplan–Meier as a nonparametric MLE

Theorem 23.4 (KM is the NPMLE, heuristic argument). *Among distributions assigning probability mass only to the observed uncensored event times, the product-limit estimator maximizes the right-censored likelihood.*

Proof sketch. At event time $y_{(j)}$, let $h_j = P(T = y_{(j)} \mid T \geq y_{(j)})$ be a discrete hazard. Among the N_j at risk, d_j fail, behaving like a local Binomial likelihood $h_j^{d_j}(1 - h_j)^{N_j - d_j}$. Maximizing the log-likelihood $d_j \log h_j + (N_j - d_j) \log(1 - h_j)$: $d_j/h_j - (N_j - d_j)/(1 - h_j) = 0 \Rightarrow \hat{h}_j = d_j/N_j$, so the estimated local survival is $1 - d_j/N_j$. Multiplying these local factors across all event times $\leq t$ gives exactly the KM estimator, i.e., KM emerges from maximizing a product of local Binomial likelihoods, one per distinct event time. $\square$

Efron's redistribution-of-mass construction

An equivalent constructive algorithm (Efron, 1967): (1) order observations, ties broken with failures before censored; (2) put mass $1/n$ at each observation; (3) sweeping left to right, whenever a censored observation is reached, redistribute its current mass equally among all observations to its right; (4) continue until all censoring mass is redistributed (if the largest observation is censored, its mass remains beyond the largest observed time). Censoring does not destroy probability mass – it says “the failure happened somewhere later,” and Efron’s algorithm literally moves that mass rightward.

23.4 Nelson–Aalen estimator

With $Y(v_j)$ the risk-set size at event time v_j (same as N_j), the Nelson–Aalen estimator of cumulative hazard is $\hat{H}_{\text{NA}}(t) = \sum_{j: v_j \leq t} d_j / Y(v_j)$, with associated survival estimate $\hat{S}_{\text{NA}}(t) = \exp\{-\hat{H}_{\text{NA}}(t)\}$. KM *multiplies* local survival factors $\prod (1 - d_j / Y_j)$; Nelson–Aalen *adds* local hazard increments then exponentiates the negative sum. The two nearly agree when each d_j / Y_j is small (since $1 - x \approx e^{-x}$), but can diverge when risk sets are small late in follow-up.

23.5 Median survival, goodness-of-fit, and uncertainty

The **median survival** is $\hat{m} = \inf\{t : \hat{S}(t) \leq 0.5\}$ – the first time the KM step function reaches or crosses 0.5.

Graphical goodness-of-fit. Substituting $\hat{S}(t)$ for $S(t)$ in a transformation that should be linear under a candidate parametric family gives a visual diagnostic (not a formal test):

- **Exponential:** $S(t) = e^{-\lambda t} \Rightarrow \log S(t) = -\lambda t$ – plot $\log \hat{S}(t)$ vs. t ; should look linear through the origin with slope $-\lambda$.
- **Weibull:** $S(t) = \exp\{-(\lambda t)^p\} \Rightarrow \log[-\log S(t)] = p \log \lambda + p \log t$ – plot $\log[-\log \hat{S}(t)]$ vs. $\log t$; should look linear with slope $\approx p$.

Greenwood’s formula. A standard large-sample variance estimate is

$$\widehat{\text{Var}}\{\hat{S}(t)\} \approx \hat{S}(t)^2 \sum_{j: y_{(j)} \leq t} \frac{d_j}{N_j(N_j - d_j)}.$$

Remark 23.5 (Heuristic derivation). Since $\log \hat{S}(t) = \sum_j \log(1 - d_j / N_j)$ and each local d_j behaves approximately as Binomial(N_j, h_j) with $\text{Var}(\hat{h}_j) \approx h_j(1 - h_j) / N_j$, a delta-method / independent-increments argument applied to $\log(1 - \hat{h}_j)$ gives $\text{Var}\{\log(1 - \hat{h}_j)\} \approx d_j / \{N_j(N_j - d_j)\}$; summing these (approximately independent) log-scale variances across event times and mapping back through $\hat{S}(t) = \exp\{\sum_j \log(1 - \hat{h}_j)\}$ via the delta method yields Greenwood’s formula above. The variance accumulates over event times and increases sharply when risk sets become small – so the KM curve is most reliable where a reasonable number of subjects remain at risk, and its right tail should be interpreted cautiously.

Kaplan–Meier reporting checklist

Number at risk at selected times; number of events; number censored; median survival (if estimable); confidence intervals for survival probabilities or median; and an explicit statement of whether censoring is plausibly independent/noninformative.

24 From Kaplan–Meier to Cox Regression

24.1 Why Kaplan–Meier is not enough

KM excellently estimates one survival curve or compares a few groups, but: it does not naturally handle continuous covariates; splitting a continuous covariate into groups loses information and creates arbitrary

cutpoints; with several covariates, stratified KM curves create many sparse cells (e.g., 2 treatment groups $\times$ 3 age groups $\times$ 4 stages = 24 strata); and it gives no compact per-covariate effect summary (“per 10-year increase in age”). We need *regression*.

Group comparison	Regression modelling
“Do survival curves differ between groups?”	“How does survival change with covariates?”
Usually categorical groups	Categorical <i>and</i> continuous covariates
KM curves, log-rank test	Cox model, parametric regression
Mostly unadjusted	Can adjust for multiple covariates

For an ordinary continuous response the analogy is: group means $\rightarrow$ linear regression $\mathbb{E}(Y | X) = x^\top \beta$. For survival, the analogous modelling target is the **hazard function** $h(t | x)$ – it is local in time (focuses on individuals who have survived to t), lets covariates act on the instantaneous event rate, and yields the interpretable **hazard ratio**.

24.2 The proportional hazards idea

Definition 24.1. Hazards are **proportional** for covariate values x_a, x_b if $h(t | x_a)/h(t | x_b)$ does not depend on t .

A proportional-hazards model is *multiplicative*: $h(t | x) = h_0(t) \times$ (relative risk depending on x). E.g., a treatment reducing hazard by 30% means $h(t | \text{new}) = 0.70 h(t | \text{control})$ for *all* t – a statement about instantaneous rates, not about survival probabilities being 30% larger at every time.

24.3 The Cox proportional hazards model

Definition 24.2 (Cox model, Cox 1972).

$$h(t | x) = h_0(t) \exp(x^\top \beta),$$

where $h_0(t)$ is an unspecified **baseline hazard** and β the covariate-effect vector.

If $x = 0$, $h(t | 0) = h_0(t)$: the baseline hazard is exactly the hazard for the (possibly centered) reference covariate profile. The exponential link keeps the hazard nonnegative for any real linear predictor, makes effects additive on the log-hazard scale ($\log h(t | x) = \log h_0(t) + x^\top \beta$), and gives coefficients simple hazard-ratio interpretations. Because $h_0(\cdot)$ is left unspecified, the Cox model is **semiparametric**: parametric in the covariate effect, nonparametric in the baseline hazard.

Proposition 24.3 (The Cox model implies proportional hazards). *For any x_a, x_b , $\frac{h(t | x_a)}{h(t | x_b)} = \exp\{(x_a - x_b)^\top \beta\}$, independent of t .*

Proof. $\frac{h(t | x_a)}{h(t | x_b)} = \frac{h_0(t) \exp(x_a^\top \beta)}{h_0(t) \exp(x_b^\top \beta)} = \exp\{(x_a - x_b)^\top \beta\}$; the unspecified $h_0(t)$ cancels exactly, and the ratio does not involve t . $\square$

Proposition 24.4 (Implied survival curves). *Let $H_0(t) = \int_0^t h_0(u) du$, $S_0(t) = \exp\{-H_0(t)\}$. Then $S(t | x) = S_0(t)^{\exp(x^\top \beta)}$.*

Proof.

$$H(t | x) = \int_0^t h_0(u) \exp(x^\top \beta) du = \exp(x^\top \beta) \int_0^t h_0(u) du = H_0(t) \exp(x^\top \beta),$$

since $\exp(x^\top \beta)$ does not depend on u . Hence

$$S(t | x) = \exp\{-H(t | x)\} = \exp\{-H_0(t) \exp(x^\top \beta)\} = (\exp\{-H_0(t)\})^{\exp(x^\top \beta)} = S_0(t)^{\exp(x^\top \beta)}. \quad \square$$

The Cox model thus generates an entire family of survival curves by raising one baseline curve $S_0(t)$ to a covariate-dependent power; these curves are generally *not* parallel – the proportionality lives on the hazard scale, not the survival scale, and under a genuine constant hazard ratio, $S_0(t)^{\text{HR}}$ curves for different groups can converge toward 0 but never cross.

24.4 The hazard ratio

For a one-unit increase in x_k (others fixed), $\text{HR}_k = e^{\beta_k}$: $\text{HR} = 1$ no difference, > 1 higher instantaneous rate, < 1 lower. More generally, for two covariate profiles, $\text{HR}(a : b) = \exp\{(x_a - x_b)^\top \beta\}$.

Example 24.5. If $\hat{\beta}_{\text{treat}} = -0.36$, $\hat{\beta}_{\text{age10}} = 0.30$ (per decade), comparing patient A (new treatment, age 70) to patient B (control, age 60): $x_a - x_b = (1, 1)^\top$, so $\log \text{HR}(A:B) = -0.36 + 0.30 = -0.06 \Rightarrow \text{HR} = e^{-0.06} = 0.94$ – the treatment benefit and age disadvantage nearly cancel.

A common misinterpretation

$\text{HR} = 0.60$ for treatment does *not* mean “40% higher survival probability at every time.” It means: at each t , among those still at risk just before t , the treated group has 40% lower instantaneous event rate than control, under proportional hazards. The gap in survival *probabilities* depends on t and on the unspecified $h_0(t)$ – it is not simply “40%” everywhere. For continuous covariates, always state the unit explicitly (1 year vs. 10 years vs. 1 SD), since the same $\hat{\beta}$ looks very different on different scales.

24.5 Risk sets and the partial likelihood

Cox estimation reuses the Kaplan–Meier risk set $R(t_j) = \{i : Y_i \geq t_j\}$. At event time t_j , suppose exactly one individual i fails.

Proposition 24.6 (Partial-likelihood contribution). *Under the Cox model, the contribution of t_j to Cox’s partial likelihood is*

$$\frac{\exp(x_i^\top \beta)}{\sum_{\ell \in R(t_j)} \exp(x_\ell^\top \beta)},$$

free of the unknown baseline hazard $h_0(\cdot)$.

Heuristic derivation. Each $\ell \in R(t_j)$ has instantaneous hazard $h_\ell(t_j) = h_0(t_j) \exp(x_\ell^\top \beta)$. For small Δt , the (approximate) probability that, given exactly one failure in $R(t_j)$ at t_j , it is specifically subject i is proportional to i ’s hazard relative to the sum of hazards of everyone at risk:

$$P(i \text{ fails} \mid \text{one failure in } R(t_j) \text{ at } t_j) \approx \frac{h_i(t_j) \Delta t}{\sum_{\ell \in R(t_j)} h_\ell(t_j) \Delta t} = \frac{h_0(t_j) \exp(x_i^\top \beta)}{\sum_{\ell} h_0(t_j) \exp(x_\ell^\top \beta)} = \frac{\exp(x_i^\top \beta)}{\sum_{\ell \in R(t_j)} \exp(x_\ell^\top \beta)},$$

the common factor $h_0(t_j)$ cancelling top and bottom. $\square$

Multiplying these contributions over all event times t_j (with $i(j)$ the individual failing at t_j) gives Cox’s partial likelihood

$$L(\beta) = \prod_j \frac{\exp(x_{i(j)}^\top \beta)}{\sum_{\ell \in R(t_j)} \exp(x_\ell^\top \beta)}.$$

Because $h_0(\cdot)$ never appears, β can be estimated by maximizing $L(\beta)$ without ever specifying $h_0(t)$ – the heart of the semiparametric Cox approach.

Handling tied event times

When more than one individual fails at the same t_j , the single-failure argument above needs modification. Two standard approximations (full derivation beyond this course): **Breslow’s approximation** treats the tied failures’ combined risk-set contribution as a simple product over the tied individuals using the same risk-set denominator; **Efron’s approximation** is more accurate, progressively reducing the risk-set sum as each tied failure is “used up.” Efron’s method is the usual software default when ties are non-negligible.

24.6 Examples across domains

- **Clinical trial:** $h(t | x) = h_0(t) \exp(\beta_1 \text{Trt} + \beta_2 \text{Age10} + \beta_3 \text{StageIII} + \beta_4 \text{StageIV})$ – the treatment hazard ratio e^{β_1} is adjusted for age and stage.
- **Reliability engineering:** $h(t | x) = h_0(t) \exp(\beta_1 \text{Temp} + \beta_2 \text{Vibration} + \beta_3 \text{Load})$; $e^{\beta_1} = 1.25$ per 10° means each 10° rise is associated with 25% higher failure hazard, other covariates fixed.
- **Customer churn:** time until a customer leaves a service, covariates price/support tickets/usage/plan/region; censoring means the customer was still active when the dataset closed. Survival methods apply whenever the response is a time-to-event, not only in medicine.

24.7 Assumptions and diagnostics

Assumption	Meaning
Proportional hazards	covariate hazard ratios are constant over time
Log-linear covariate effect	$x^\top \beta$ is an appropriate model on the log-hazard scale
Independent censoring	censoring is not informative after conditioning on covariates

Proportional hazards does *not* mean survival curves are parallel, that survival differences are constant over time, that median survival times have a fixed ratio, or that any estimated effect is automatically causal. It means precisely that $h(t | x_a)/h(t | x_b)$ is constant in t . The assumption may fail when risk changes shape over time (e.g., high early surgical risk but long-term benefit, or a drug effect that wears off) – a visual warning sign is substantially **crossing** KM curves between groups, since true proportional-hazards survival curves $S_0(t)^{\text{HR}}$ can converge but never cross. Remedies (beyond this course) include time-varying coefficients, stratified Cox models, and non-PH alternatives.

KM still matters after fitting a Cox model

KM curves remain useful for exploratory plots by major groups, informal visual PH diagnostics (crossing curves), communicating unadjusted survival experience to non-technical audiences, and comparison against model-based covariate-adjusted curves. A good survival analysis typically uses both.

24.8 Comparing groups formally: the log-rank test (preview)

Before fitting a Cox model, a common first step is to test whether two (or more) KM curves differ using the **log-rank test**: at each observed event time t_j , compare the *observed* number of events in each group to the number *expected* under the null hypothesis of equal hazards, using the same hypergeometric risk-set logic as a 2×2 table at each t_j ; summing observed-minus-expected counts (and their variances) across event times gives a chi-squared test statistic. This is a special case (no covariates, no proportional-hazards parametrization) of the same risk-set machinery that powers both Kaplan–Meier and the Cox partial likelihood – full derivation and worked examples are left to the next lecture on inference for the Cox model.

24.9 Concept map

$$\underbrace{\text{Kaplan–Meier risk sets}}_{R(t_j)} \longrightarrow \underbrace{\text{Hazard}}_{h(t|x), \text{ conditional rate}} \longrightarrow \underbrace{\text{Cox model}}_{h_0(t)e^{x^\top \beta}} \longrightarrow \underbrace{\text{Hazard ratios}}_{e^{\beta_k}}$$

The Cox model is best understood as *risk-set-based regression* for event times: it reuses the risk-set machinery of Kaplan–Meier but replaces simple per-group counts with a regression-weighted comparison via $\exp(x^\top \beta)$.

What comes next

Full derivation of the partial likelihood and its score/information; formal tie-breaking (Breslow/Efron); Wald, score, and likelihood-ratio tests for covariate effects; Schoenfeld residuals for checking proportional hazards; and constructing adjusted survival curves from a fitted Cox model.

Overall Exam Checklist

Master checklist across all lectures

- **EM:** distinguish θ (optimized) from $\theta^{(t)}$ (fixes the E-step distribution); never skip the constraint's Lagrange multiplier; use the *updated* mean in variance updates; EM ascent is guaranteed, global optimality is not.
- **Mixtures:** responsibilities are soft (fractional) labels, not hard assignments; watch for label switching and the identical-components unidentifiability trap.
- **Incomplete tables:** a single category can receive expected contributions from multiple ambiguous groups; the M-step is always “normalize the expected completed counts.”
- **MM:** majorize-then-minimize (loss) or minorize-then-maximize (likelihood); tangency at $\theta^{(t)}$ plus domination/dominance-reversal is what buys the descent/ascent guarantee, even under inexact (generalized) M-steps.
- **Survival:** a censored time is an inequality, not a missing value – never delete censored records or treat them as exact events; hazard is a rate, not a probability; KM multiplies local survival factors over risk sets and is the NPMLE.
- **Cox:** hazard ratios are constant-over-time *rate* ratios, not survival-probability ratios; the partial likelihood cancels the unknown baseline hazard exactly because it is common to every member of a risk set.